

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**  
**Faculty of Technology & Engineering**  
**Devang Patel Institute of Advance Technology and Research (DEPSTAR)**

**CSUE301 – : Big Data Analytics**  
**Practical File – ODD Semester 2026-27**

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## Practical 9

### 1. Problem Definition

TrendPulse Social Media Analytics monitors social media activity across platforms such as Twitter/X, Instagram, Facebook, and LinkedIn. Large volumes of posts contain hashtags related to technology, sports, entertainment and other topics.

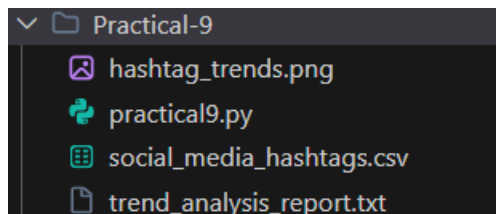
The objective of this practical is to use **PySpark** to process hashtag data and identify popular and trending topics.

The practical performs hashtag grouping, frequency counting, popularity ranking, daily trend analysis, regional analysis, category analysis and visualization.

**Note:** The dataset used in this practical is a generated/sample dataset for educational analysis.

### PART A — Practical Execution

#### Step 1: Creating Folder and files



#### Step 2: Creating Spark Session

```
=====
TASK 1: CREATING SPARK SESSION
=====
WARNING: Using incubator modules: jdk.incubator.vector
26/09/27 17:08:49 WARN Shell: Did not find winutils.exe: java.io.FileNotFoundException: java.io.FileNotFoundException: HADOOP_HOME and hadoop.home.dir are unset. -see
https://wiki.apache.org/confluence/display/HADOOP2/WindowsProblems
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/09/27 17:08:50 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Spark Version      : 4.2.0
Spark App Name     : Social-Media-Trend-Detection
Master URL         : local[*]
```

**Step 3: Loading Social Media Hashtag Dataset**

## ===== TASK 2: LOADING SOCIAL MEDIA HASHTAG DATASET =====

Dataset Path : C:\Users\ADMIN\OneDrive\Desktop\BDA\Practical-9\social\_media\_hashtags.csv

Dataset loaded successfully.

Total Number of Records: 650

First 10 Records:

| Timestamp  | Platform  | Hashtag         | Region        |
|------------|-----------|-----------------|---------------|
| 2026-03-04 | LinkedIn  | #Netflix        | Latin America |
| 2026-03-04 | Facebook  | #CloudComputing | North America |
| 2026-03-01 | Instagram | #Netflix        | Asia-Pacific  |
| 2026-03-04 | Facebook  | #CloudComputing | Middle East   |
| 2026-03-01 | LinkedIn  | #Tech           | Middle East   |
| 2026-03-01 | LinkedIn  | #Movies         | Europe        |
| 2026-03-01 | LinkedIn  | #Hollywood      | Europe        |
| 2026-03-04 | LinkedIn  | #Netflix        | North America |
| 2026-03-04 | LinkedIn  | #AI             | North America |
| 2026-03-05 | Twitter/X | #AI             | North America |

only showing top 10 rows

**Step 4: Displaying Hashtag Records**

## ===== TASK 3: DISPLAYING HASHTAG RECORDS =====

Sample Hashtag Records (First 10 rows):

| Timestamp  | Platform  | Hashtag         | Region        |
|------------|-----------|-----------------|---------------|
| 2026-03-04 | LinkedIn  | #Netflix        | Latin America |
| 2026-03-04 | Facebook  | #CloudComputing | North America |
| 2026-03-01 | Instagram | #Netflix        | Asia-Pacific  |
| 2026-03-04 | Facebook  | #CloudComputing | Middle East   |
| 2026-03-01 | LinkedIn  | #Tech           | Middle East   |
| 2026-03-01 | LinkedIn  | #Movies         | Europe        |
| 2026-03-01 | LinkedIn  | #Hollywood      | Europe        |
| 2026-03-04 | LinkedIn  | #Netflix        | North America |
| 2026-03-04 | LinkedIn  | #AI             | North America |
| 2026-03-05 | Twitter/X | #AI             | North America |

only showing top 10 rows

**Step 5: Validating Dataset Structure**

```
=====
TASK 4: VALIDATING DATASET STRUCTURE
=====
DataFrame Schema:
root
|-- Timestamp: date (nullable = true)
|-- Platform: string (nullable = true)
|-- Hashtag: string (nullable = true)
|-- Region: string (nullable = true)

Number of Columns : 4
Number of Records : 650
Column Names      : Timestamp, Platform, Hashtag, Region
```

**Step 6: Grouping Hashtags**

```
=====
TASK 5: GROUPING HASHTAGS
=====
Hashtag Occurrences Grouped (Top 15):
+-----+-----+
|Hashtag      |count|
+-----+-----+
|#AI          |75   |
|#Cricket     |68   |
|#MachineLearning|58   |
|#Netflix     |52   |
|#DataScience |45   |
|#Football    |42   |
|#Gaming      |38   |
|#Tech        |35   |
|#IPL         |34   |
|#CloudComputing|32   |
|#CyberSecurity|28   |
|#Movies      |26   |
|#Music       |24   |
|#Sports      |22   |
|#Bollywood   |20   |
+-----+-----+
only showing top 15 rows
```

**Step 7: Counting Hashtag Occurrences**

```
=====
TASK 6: COUNTING HASHTAG OCCURRENCES
=====
```

Top 10 Most Frequent Hashtags:

| Hashtag          | count |
|------------------|-------|
| #AI              | 75    |
| #Cricket         | 68    |
| #MachineLearning | 58    |
| #Netflix         | 52    |
| #DataScience     | 45    |
| #Football        | 42    |
| #Gaming          | 38    |
| #Tech            | 35    |
| #IPL             | 34    |
| #CloudComputing  | 32    |

**Step 8: Sorting Hashtags by Popularity**

```
=====
TASK 7: SORTING HASHTAGS BY POPULARITY
=====
```

Hashtags Sorted by Popularity with Dynamic Rank:

| Rank | Hashtag          | Occurrence_Count |
|------|------------------|------------------|
| 1    | #AI              | 75               |
| 2    | #Cricket         | 68               |
| 3    | #MachineLearning | 58               |
| 4    | #Netflix         | 52               |
| 5    | #DataScience     | 45               |
| 6    | #Football        | 42               |
| 7    | #Gaming          | 38               |
| 8    | #Tech            | 35               |
| 9    | #IPL             | 34               |
| 10   | #CloudComputing  | 32               |
| 11   | #CyberSecurity   | 28               |
| 12   | #Movies          | 26               |
| 13   | #Music           | 24               |
| 14   | #Sports          | 22               |
| 15   | #Bollywood       | 20               |

only showing top 15 rows

**Step 9: Identifying Top Trending Hashtags**

## ===== TASK 8: IDENTIFYING TOP TRENDING HASHTAGS =====

Top 5 Trending Hashtags:

| Rank | Hashtag          | Occurrence_Count |
|------|------------------|------------------|
| 1    | #AI              | 75               |
| 2    | #Cricket         | 68               |
| 3    | #MachineLearning | 58               |
| 4    | #Netflix         | 52               |
| 5    | #DataScience     | 45               |

[&gt;&gt;&gt;] Top Trending Hashtag: #AI with 75 posts

**Step 10: Supplementary 1 — Top 3 Trending Hashtags**

## ===== SUPPLEMENTARY 1: TOP 3 TRENDING HASHTAGS =====

| Rank | Hashtag          | Occurrence_Count |
|------|------------------|------------------|
| 1    | #AI              | 75               |
| 2    | #Cricket         | 68               |
| 3    | #MachineLearning | 58               |

Top Trending Hashtag:
#AI (Total Occurrences: 75)

**Step 11: Supplementary 2 — Daily Trend Comparison**

## ===== SUPPLEMENTARY 2: DAILY TREND COMPARISON =====

Comparing Hashtag Trends Across Multiple Days...

Daily Hashtag Counts (Sample 12 records):

| Timestamp  | Hashtag          | count |
|------------|------------------|-------|
| 2026-03-01 | #AI              | 16    |
| 2026-03-01 | #Netflix         | 12    |
| 2026-03-01 | #Cricket         | 11    |
| 2026-03-01 | #DataScience     | 10    |
| 2026-03-01 | #Movies          | 9     |
| 2026-03-01 | #IPL             | 8     |
| 2026-03-01 | #CloudComputing  | 7     |
| 2026-03-01 | #Bollywood       | 7     |
| 2026-03-01 | #Tech            | 7     |
| 2026-03-01 | #Gaming          | 6     |
| 2026-03-01 | #Football        | 6     |
| 2026-03-01 | #MachineLearning | 6     |

only showing top 12 rows

Most Popular Hashtag Per Day:

| Timestamp  | Daily_Top_Hashtag | Daily_Count |
|------------|-------------------|-------------|
| 2026-03-01 | #AI               | 16          |
| 2026-03-02 | #AI               | 17          |
| 2026-03-03 | #Cricket          | 16          |
| 2026-03-04 | #AI               | 19          |
| 2026-03-05 | #Cricket          | 14          |

**Step 12: Supplementary 3 — Regional Hashtag Analysis**

```

=====
SUPPLEMENTARY 3: REGIONAL HASHTAG ANALYSIS
=====

Total Posts Per Region:
+-----+-----+
|Region      |Total_Posts|
+-----+-----+
|Asia-Pacific|145        |
|North America|142       |
|Middle East |122       |
|Europe      |121       |
|Latin America|120      |
+-----+-----+

Top Trending Hashtag Per Region:
+-----+-----+-----+
|Region      |Top_Hashtag|Post_Count|
+-----+-----+-----+
|Asia-Pacific|#Cricket    |20        |
|Latin America|#MachineLearning|20       |
|North America|#AI         |20        |
|Middle East  |#AI         |16        |
|Europe       |#MachineLearning|15       |
+-----+-----+-----+

```

**Step 13: Task 10 / Supplementary 4 — User Interest and Hashtag Categories**

```

=====
TASK 10 & SUPPLEMENTARY 4: USER INTEREST & HASHTAG CATEGORIES
=====

User Interest Distribution by Category:
+-----+-----+
|Category      |Number_of_Posts|
+-----+-----+
|Technology     |273            |
|Sports         |199            |
|Entertainment  |178            |
+-----+-----+

[>>>] Most Popular Category: Technology with 273 posts

Platform Distribution:
+-----+-----+
|Platform |Total_Posts|
+-----+-----+
|Facebook |172        |
|Instagram|165        |
|LinkedIn |160        |
|Twitter/X|153        |
+-----+-----+

```

## Step 14: Supplementary 5 — Trend Visualization

### SUPPLEMENTARY 5: TREND VISUALIZATION

Creating Matplotlib Bar Chart for Top 10 Trending Hashtags...

Chart successfully saved to: C:\Users\ADMIN\OneDrive\Desktop\BDA\Practical-9\hashtag\_trends.png

### BUSINESS RECOMMENDATIONS

[DISCLAIMER]: Educational analysis based on generated/sample social media data.

1. Campaign Targeting : Allocate top-tier marketing budgets toward Technology and Sports hashtags.
2. Platform Strategy : Cross-post top tech trends on LinkedIn and fan-driven sports topics on Twitter/X.
3. Emerging Topics : Monitor real-time daily trend shifts to detect breakout virality early.
4. Geo-Targeted Content : Tailor campaigns based on regional preferences (e.g. cricket in Asia-Pacific).
5. Community Engagement : Create branded interactive polls around #AI, #Cricket, and #Netflix.

### FINAL SOCIAL MEDIA TREND ANALYSIS REPORT

#### TRENDPULSE SOCIAL MEDIA TREND DETECTION REPORT

#### 1. DATASET DETAILS:

- Source File : social\_media\_hashtags.csv
- Total Records 650
- Schema : Timestamp (string), Platform (string), Hashtag (string), Region (string)
- Data Nature : Generated / Sample dataset for educational evaluation

#### 2. TOTAL RECORDS:

- Total Monitored Posts 650

#### 3. UNIQUE HASHTAGS:

- Total Distinct Hashtags 18

#### 4. PLATFORM DISTRIBUTION:

| Platform | Total Posts |
|----------|-------------|
|----------|-------------|

|           |     |
|-----------|-----|
| Facebook  | 172 |
| Instagram | 165 |
| LinkedIn  | 160 |

Twitter/X | 153

#### 5. REGION DISTRIBUTION:

| Region        | Top Hashtag      | Post Count |
|---------------|------------------|------------|
| -----         |                  |            |
| Asia-Pacific  | #Cricket         | 20         |
| Latin America | #MachineLearning | 20         |
| North America | #AI              | 20         |
| Middle East   | #AI              | 16         |
| Europe        | #MachineLearning | 15         |

#### 6. HASHTAG FREQUENCY ANALYSIS:

- Evaluated across all monitored interactions with dynamic grouping and frequency sorting.

#### 7. TOP 10 HASHTAGS:

| Rank  | Hashtag          | Occurrences |
|-------|------------------|-------------|
| ----- |                  |             |
| 1     | #AI              | 75          |
| 2     | #Cricket         | 68          |
| 3     | #MachineLearning | 58          |
| 4     | #Netflix         | 52          |
| 5     | #DataScience     | 45          |
| 6     | #Football        | 42          |
| 7     | #Gaming          | 38          |
| 8     | #Tech            | 35          |
| 9     | #IPL             | 34          |
| 10    | #CloudComputing  | 32          |

#### 8. TOP 5 TRENDING HASHTAGS:

| Rank  | Hashtag          | Occurrences |
|-------|------------------|-------------|
| ----- |                  |             |
| 1     | #AI              | 75          |
| 2     | #Cricket         | 68          |
| 3     | #MachineLearning | 58          |
| 4     | #Netflix         | 52          |
| 5     | #DataScience     | 45          |

#### 9. TOP 3 TRENDING HASHTAGS:

| Rank  | Hashtag          | Occurrences |
|-------|------------------|-------------|
| ----- |                  |             |
| 1     | #AI              | 75          |
| 2     | #Cricket         | 68          |
| 3     | #MachineLearning | 58          |

\* Top Trending Hashtag : #AI (75 occurrences)

## 10. DAILY TREND ANALYSIS:

| Date       | Daily Top Hashtag | Post Count |
|------------|-------------------|------------|
| -----      |                   |            |
| 2026-03-01 | #AI               | 16         |
| 2026-03-02 | #AI               | 17         |
| 2026-03-03 | #Cricket          | 16         |
| 2026-03-04 | #AI               | 19         |
| 2026-03-05 | #Cricket          | 14         |

## 11. REGIONAL ANALYSIS:

| Region        | Top Hashtag      | Post Count |
|---------------|------------------|------------|
| -----         |                  |            |
| Asia-Pacific  | #Cricket         | 20         |
| Latin America | #MachineLearning | 20         |
| North America | #AI              | 20         |
| Middle East   | #AI              | 16         |
| Europe        | #MachineLearning | 15         |

## 12. CATEGORY ANALYSIS:

| Category      | Number of Posts |
|---------------|-----------------|
| -----         |                 |
| Technology    | 273             |
| Sports        | 199             |
| Entertainment | 178             |

\* Leading Category : Technology (273 posts)

## 13. USER INTEREST ANALYSIS:

- User attention is primarily concentrated in cutting-edge Technology topics (#AI, #MachineLearning), followed by high-energy Sports discussions (#Cricket, #Football) and Entertainment streaming (#Netflix).

## 14. BUSINESS RECOMMENDATIONS:

- \* DISCLAIMER: Educational analysis based on generated/sample social media data.
- Allocate promotional spending to high-affinity topics (#AI, #Cricket, #MachineLearning).
- Leverage multi-platform distribution based on user demographics (Tech -> LinkedIn, Sports -> Twitter/X).
- Monitor daily breakout hashtags to optimize real-time digital marketing responses.

## 15. CONCLUSION:

PySpark DataFrame transformations, grouping, and window functions offer massive scalability and low latency for parsing unstructured social interaction streams and extracting actionable trends.

=====

**Key Questions / Analysis****Q1. Why is Apache Spark suitable for social media analytics? Answer:**

Apache Spark is suitable for social media analytics because it can process large amounts of data efficiently using distributed computing. Spark DataFrames provide fast operations such as filtering, grouping, aggregation and sorting, which are useful for analysing large social media datasets.

**Q2. What is the significance of hashtag frequency?****Answer:**

Hashtag frequency represents how many times a particular hashtag appears in the dataset. A higher frequency indicates that the topic is receiving more attention in the analyzed sample.

**Q3. How can trend analysis support digital marketing campaigns? Answer:**

Trend analysis helps marketers identify popular topics and changing user interests. Businesses can use this information to create relevant content, plan campaigns and monitor emerging topics.

**Q4. Why do businesses monitor social media trends? Answer:**

Businesses monitor social media trends to understand customer interests, identify emerging topics, observe discussions about products or services and improve their marketing strategies.

**Q5. How can real-time analytics improve decision making? Answer:**

Real-time analytics allows organizations to detect changes in trends quickly. This can help businesses respond faster to emerging topics, customer interests and market changes.

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## Result

The Social Media Trend Detection practical was successfully implemented using PySpark.

The dataset contained 650 social media records and 18 distinct hashtags. Hashtag frequency, popularity ranking, top trends, daily trends, regional trends and category-based user interests were successfully analyzed.

The most frequently occurring hashtag in the sample dataset was #AI with 75 occurrences.

The Technology category had the highest number of posts with 273 posts. A visualization of the top 10 hashtags was also successfully generated using Matplotlib.

## Conclusion

The practical demonstrated how **PySpark DataFrame operations and aggregation techniques** can be used for social media trend detection. Hashtags were grouped, counted and ranked according to their frequency.

Daily and regional analysis provided additional information about changing user interests. Category analysis showed the distribution of Technology, Sports and Entertainment topics.

The practical demonstrates that PySpark can be used to process and analyze large-scale social media datasets efficiently and extract useful trends for business analysis.

## 5. Github Link

Github Link: [24DIT041\\_BDA/Practical-9 at main · Krina-14/24DIT041\\_BDA](#)